LESSON PLAN

FOLG (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Engine Health monitoring by using SOAP Analysis
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Engine Health monitoring by using SOAP Analysis
5. Specific Objectives	26.1.1.3 Trainee will be able to explain the SOAP Analysis EHM Technique

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of SOAP logbook entries; Safety: Handle logbooks carefully; Supervision: Instructor monitors logbook analysis
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What does EHM stand for?
(b) List the four components of SOAP analysis.
(c) Why is engine health monitoring critical for aircrew?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Engine Health Monitoring • Purpose of SOAP Analysis technique • Overview of S-O-A-P acronym • Role of EHM in maintenance planning • Importance of accurate data recording	L & Ds	PPT, IB	15 min
(b)	• Detailed explanation of 'S' in SOAP • Detailed explanation of 'O' in SOAP • Detailed explanation of 'A' in SOAP • Detailed explanation of 'P' in SOAP • Interpreting trends in engine parameters	L & Ds	PPT, PDF	20 min
(c)	• Case study of SOAP analysis application • Common errors in SOAP data entry • Impact of SOAP on flight safety • Integration with other monitoring systems • Best practices for logbook maintenance	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the significance of the 'S' component in SOAP analysis.
(b) How does SOAP analysis contribute to predictive maintenance?
(c) Identify one common error in recording SOAP data.

12. Home Assignment and Follow Up	Prepare a brief summary of how SOAP analysis bridges the gap between current monitoring and the debris analysis techniques covered in LP 05.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Engine Health monitoring by using Debris analysis.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min